Haniyeh Zahra Budaqi

Email | LinkedIn | GitHub | Google Scholar

Mechanical Engineering graduate with research experience in cislunar astrodynamics, trajectory generation, and spacecraft dynamics and control. Interested in space systems design, mission analysis, and computational methods for future space missions

Education

B.Sc. Mechanical Engineering Nov 2020 – Jul 2025
Sharif University of Technology *Tehran, Iran*

- **GPA:** 16.69/20.00
- **Focus Areas:** Dynamics, Robotics, Control Systems
- **Relevant Coursework:** Modern Control Theory, Robotics, Orbital Mechanics
- **Thesis:** "Postural Responses to Mediolateral Rotations," supervised by Dr. Saeed Behzadipour

Research Focus

- Cislunar astrodynamics, periodic orbits, and trajectory generation in the Earth–Moon three-body problem
- Space mission analysis, trajectory optimization, and computational methods for space systems design
- Spacecraft dynamics and control, with interest in autonomous and robust mission operations

Publications

"Projection-Constrained Generative Initialization of NRHO-DRO Transfers in the Earth-Moon CR3BP" Sep 2026
H. Z. Budaqi, *P. Latifiyan, M. Arsalan* *Submitted to Acta Astronautica*
View Manuscript

- Developed cVAE-projected, a projection-constrained conditional VAE generating initial-guess NRHO–DRO transfers in the Earth–Moon CR3BP, trained on 57,356 corrected trajectories.
- Achieved 80% guess-level success and 99.6%+ pair-level coverage across 3,133 held-out orbit pairs, outperforming Lambert and Poincaré-mapping baselines in coverage and compute cost.

"Postural Responses to Mediolateral Rotations" Nov 2025
H. Z. Budaqi, *A. Mojibi, S. Behzadipour* *ICBME 2025 Conference*
10.1109/ICBME68496.2025.11392127

- Derived from undergraduate thesis; designed and conducted experiments with 15 participants across eight sensory and cognitive conditions, analyzing center-of-pressure (COP) responses via repeated-measures ANOVA.
- Demonstrated that vision, surface compliance, and cognitive load independently influence postural stability during mediolateral disturbances.

Teaching Assistance

Sharif University of Technology *Tehran, Iran*

Engineering Mechanics Dynamics
Instructor: Dr. Somaye Mohammadi Fall 2026

- Lead supplementary MATLAB tutoring sessions.

Orbital Mechanics
Instructor: Dr. Maryam Malekzadeh Fall 2026

- Assign and grade problem sets.

Numerical Calculations
Instructor: Dr. Amir Nourani Fall 2022, Fall 2023

- Held MATLAB tutoring sessions and graded MATLAB assignments.

Selected Projects

Reinforcement Learning for Cislunar Mission Management | **Python, PPO** Ongoing

– Developing a Proximal Policy Optimization (PPO)-based reinforcement learning framework to support mission management decision-making in cislunar mission design; manuscript in preparation for journal submission.

Attitude Dynamics and Control of a Nano-Satellite Orbiting Mars | **MATLAB, STK** Mar 2026 – Jun 2026

– Developed a nonlinear attitude dynamics simulator and Lyapunov-based controller for a Mars-orbiting nano-satellite using Modified Rodrigues Parameters (MRPs), integrating STK visualization to assess pointing error, angular velocity, and control effort across multiple pointing modes.

Dual AC Servo Robot Control System | **Arduino, MATLAB, Python** Jul 2025

– Designed and implemented a dual AC servo-actuated robotic balance platform with a load-cell/HX711/Arduino Due data acquisition system, and developed Python/MATLAB tools for real-time perturbation generation, synchronized COP measurement, and signal processing.

Engineering Experience

Embedded Software Engineer
Lian Battery, Tehran, Iran Jan 2026 – Present

– Developing firmware systems using FreeRTOS on ESP32-S3 for real-time embedded applications.

– Designed modular software architecture for CAN-based communication systems.

MG Adak, Tehran, Iran Apr 2025 – Jan 2026

– Developed modular STM32 firmware for commercial IoT systems with focus on reliability and scalability.

– Developed Android application for Bluetooth-based data exchange.

Electronics Assistant Oct 2025 – Jan 2026
Sharif Social Robotics Lab *Tehran, Iran*

– Designed 2-layer PCBs and contributed to embedded system development for robotic platforms.

– Implemented BLE-based communication for real-time telemetry and control.

Electronics Engineering Intern Nov 2024 – Jul 2025
Nira System Pouya *Tehran, Iran*

– Passed specialized courses on Embedded C programming and PCB design, followed by hands-on projects.

Technical Skills

– **Control & Dynamics:** Spacecraft attitude dynamics, nonlinear control, PID control, system modeling, numerical simulation

– **Machine Learning:** Deep learning (conditional VAEs), reinforcement learning (PPO), applied to mission design and navigation

– **Programming & Simulation:** MATLAB/Simulink, Python, C/C++, Git

– **Embedded Systems:** STM32, ESP32, AVR; SPI, I2C, UART, DMA, CAN, BLE

– **Tools:** STK, SolidWorks, Altium Designer

– **Languages:** English (CEFR C2, IELTS Academic 8.5/9.0), French (CEFR A2, actively studying)

Courses & Certifications

– Spacecraft Dynamics and Control Specialization (CU Boulder/Coursera)

– Introduction to MATLAB Programming (Vanderbilt University/Coursera)

– Harvard University's Introduction to Computer Science (CS50)

Honors & Awards

– Ranked 97th in the National Mathematics & Physics University Entrance Exam